

Resampling Techniques

Problem 12: HPD Interval for a Skewed Posterior

Suppose the posterior distribution is:

$$\theta \mid x \sim \text{Beta}(2, 8)$$

- The posterior is skewed
- HPD interval is not equal to equal-tail interval

Goal:

- Compute and compare HPD and equal-tail intervals

Task 12.1: Posterior Simulation

- 1 Generate $N = 10,000$ samples from:

$$\theta \sim \text{Beta}(2, 8)$$

- 2 Plot:

- Histogram
- Density curve

- 3 Comment on:

- Skewness
- Location of the mode

Task 12.2: Equal-Tail Interval

Compute the 95% equal-tail interval:

$$(\theta_{0.025}, \theta_{0.975})$$

- 1 Use empirical quantiles of the simulated samples
- 2 Report the interval

Task 12.3: HPD Interval

The HPD interval is defined as:

$$(L, U) = \{\theta : \pi(\theta | x) > k\}$$

- 1 Sort the simulated samples:

$$\theta_{(1)} \leq \cdots \leq \theta_{(N)}$$

- 2 Consider all intervals of length $0.95N$
- 3 Find the interval with minimum width:

$$U - L$$

Task 12.4: Comparison and Interpretation

- ① Compare:
 - HPD interval
 - Equal-tail interval
 - Shift toward high-density region
- ② Explain why:

HPD $\neq$ Equal-tail

Problem 13: Bayesian Bootstrap

We construct a posterior using:

- Dirichlet weights
- Random weighted empirical distribution

Data is generated as:

$$x_1, \dots, x_{100} \sim \text{Exponential}(1)$$

Task 13.1: Data Simulation

- 1 Generate $n = 100$ observations
- 2 Plot histogram of the data
- 3 Comment on skewness

Task 13.2: Dirichlet Weights

Generate weights:

$$(w_1, \dots, w_{100}) \sim \text{Dirichlet}(1, \dots, 1)$$

- 1 Generate one set of weights
- 2 Verify:

$$\sum w_i = 1$$

Task 13.3: Bayesian Bootstrap Estimate

Compute the weighted mean:

$$\hat{\theta}^* = \sum_{i=1}^{100} w_i x_i$$

- 1 Repeat the procedure $B = 10,000$ times
- 2 Store all $\hat{\theta}^*$

Task 13.4: Posterior Analysis

- ① Plot histogram of $\hat{\theta}^*$
- ② Compute:
 - Posterior mean
 - Posterior variance

Task 13.5: HPD Interval

Compute the 95% HPD interval:

- ① Sort the bootstrap samples
- ② Use shortest interval method
- ③ Report (L, U)
- ④ Compare with:
 - Sample mean
 - Classical bootstrap interval